AI33-MPOPT · FIFTH PILLAR · 2026

A Geometric Transport Architecture

Symmetry, connectivity, transport, occupation and the operator algebra they force — established before any physical interpretation is assigned, and carried upward into the spectral and cosmological layers.

THE RESULT THIS FILE ESTABLISHES

The 33-domain architecture has an automorphism group of order exactly 120 with orbits {1, 12, 20}. Its admissible operators form a 23-dimensional algebra and its correlation kernel reduces to 17 parameters. Every sector-resolved quantity derived within the framework collapses to three values — proved from the structure.

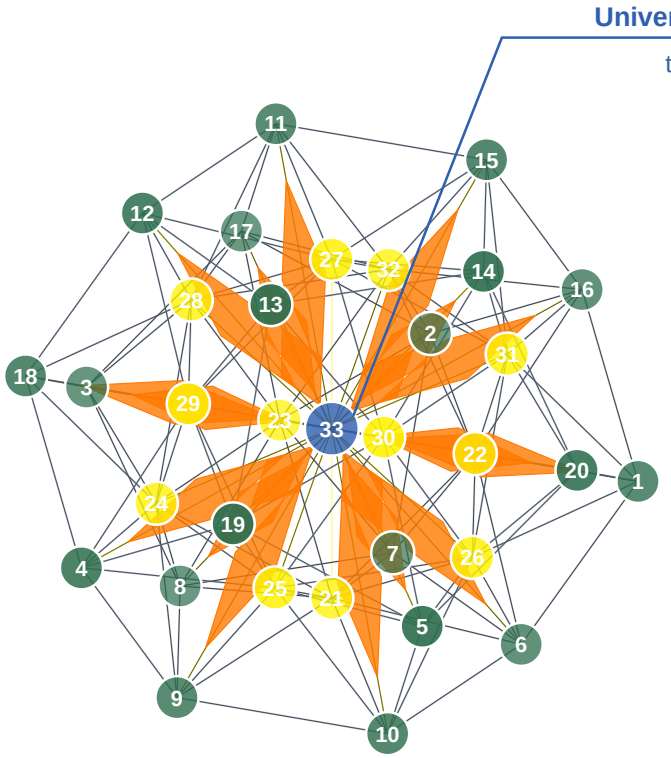
FIGURE · THE THREE ORBITS OF THE 120-ELEMENT SYMMETRY GROUP

|Aut| = 120

1	12	20
hub	inner (apex)	outer (curvilinear)

Rolando Rivero · Independent Researcher · AI33-MPOPT Framework

Companion to the Quantum-Gravity Bridge, Lawfulness, LQG Bridge and Build-Up files · for Zenodo



— 20 curvilinears C1–C20
● Hub · #33
● 12 inner · #21–32
● 20 outer · #1–20
— 32 throats to the hub
— 120 sibling links

FOUNDATION

The foundational proposal

THE QUESTION AI33-MPOPT BEGAN WITH

What if the phenomena we call dark matter and dark energy are not fundamental causes of cosmic behaviour, but observable **effects** produced by a deeper underlying architecture? Modern cosmology measures those effects precisely; their physical origin remains unresolved. This work explores an alternative: that reality may be organised by a geometric transport architecture, and that what we identify as dark matter and dark energy could be large-scale consequences of it.

SO THE INVESTIGATION DOES NOT BEGIN WITH THE DARK SECTOR

It begins with structure: **32 transport throats**, **20 curvilinear domains**, **12 apex domains**, and **1 central collecting universe**. The question becomes whether a symmetry-driven geometry can generate the transport, connectivity and occupation patterns that would later appear as physical effects.

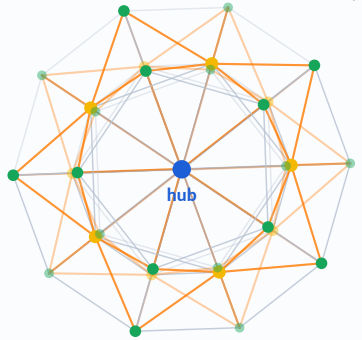
WHAT THIS FILE ESTABLISHES

That the architecture possesses a **mathematically forced** 12/20 sector structure arising from symmetry alone, and that transport, connectivity and occupation measures all inherit that partition from the geometry. These are theorems and computations, independent of any cosmological reading.

WHAT THIS WORK DOES

AI33-MPOPT proposes that dark matter and dark energy may both be consequences of a deeper symmetry architecture whose signatures can be studied mathematically. This file studies them: the structure, its theorems and its computations, and the reader judges.

FIGURE · THE 33-DOMAIN STRUCTURE THIS FILE STUDIES — 1 HUB, 12 APEX, 20 CURVILINEAR, 152 CONNECTIONS



33 DOMAINS



hub: degree 32

apex: degree 11 (5 apex, 5 curvilinear, 1 throat)

curvilinear: degree 7 (3 apex, 3 curvilinear, 1 throat)

30	60	30	32
apex–apex	apex–curvilinear	curvilinear–curvilinear	throats to the hub

= 152 connections, every count fixed by the icosahedron's vertices and faces

FRAMING

Geometry first. Observation last.

GEOMETRY

▼

SYMMETRY

▼

CONNECTIVITY

▼

TRANSPORT

▼

OCCUPATION

▼

OBSERVABLE EFFECTS

▼

DARK MATTER / DARK ENERGY ?

EVERYTHING ABOVE THE LAST TWO BOXES IS ESTABLISHED HERE

Geometry, symmetry, connectivity, transport and occupation are all computed in this file, with proofs, controls and null models. None of them requires a cosmological assumption.

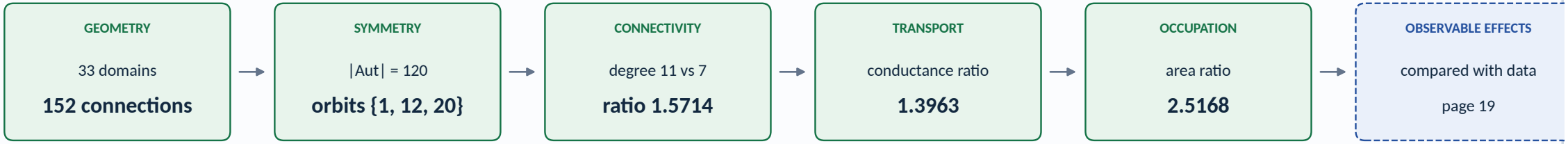
THE LAST BOX IS THE COMPARISON

Whether the architecture's occupation structure corresponds to the observed dark-sector partition is examined on page 19 — with its statistics printed alongside it. It is the final line of the chain, not the first.

WHY THE ORDER MATTERS

Standard treatments begin with matter and forces and derive geometry. This one begins with geometry and asks what must follow. Every result below the top box is a consequence, not an input.

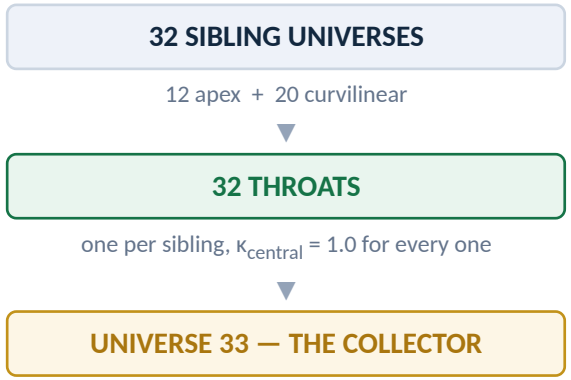
FIGURE · WHAT EACH LINK OF THE CHAIN COMPUTES



THE MECHANISM

The 32 throats are the mechanism

The 12/20 partition organises the structure. The throats carry everything through it.



THE THROATS ARE IDENTICAL. WHAT ARRIVES THROUGH THEM IS NOT.

Every throat is built with the same coupling. Yet conductance to the hub reads **9.487 at an apex against 6.635 at a curvilinear** — a 43% difference produced entirely by what surrounds each throat, not by the throat itself.

WHY: 32 THROATS AGAINST 120 SURFACE LINKS

The throats are 21% of the 152 edges; the other 120 are surface connections, shared unequally — **10 surface neighbours per apex, 6 per curvilinear**. Conductance is a whole-network property, so the surface decides what each identical throat delivers.

TWO MECHANISMS, KEPT SEPARATE

The throats transport. The 12/20 partition organises. Every result in this file is one or the other: transport quantities land near 1.4, occupation quantities near 2.5, and the two never converge.

WHAT HAPPENS IF THE THROATS ARE REMOVED

Not collapse. The 32-node shell survives as a fully connected structure on its own 120 surface edges — it is lawful without the hub. What is lost is the collector: Universe 33 disconnects entirely. And at zero throat coupling the operator becomes singular — no inverse, no Green function, no propagator, F = R = 0. The throats do not hold the shell together. They are what makes the hub exist at all.

The hub has degree 32; the next highest is 11 — not a peer node, but the single point on which the transport architecture converges.

THE DISTINCTION THIS WORK RESTS ON

POSITION

Cause and effect, reversed

CONVENTIONAL ORDER

DARK MATTER · DARK ENERGY

▼

GRAVITATIONAL BEHAVIOUR

▼

EXPANSION & STRUCTURE

Components posited first; behaviour follows. Their physical nature remains unidentified after four decades of direct-detection effort.

THE ORDER INVESTIGATED HERE

SYMMETRY ARCHITECTURE

▼

12-SECTOR / 20-SECTOR

▼

TRANSPORT & OCCUPATION

▼

OBSERVABLE PUSH / PULL ?

Geometry posited first; everything below derived. The dark sector appears at the bottom, as the result.

WHAT THIS FILE PROPOSES

The 12 apex domains are **the pull (dark matter)**; the 20 curvilinear domains are **the push (dark energy)**. The 12/20 split is proved from the geometry, as Jacobson (1995) derived the Einstein equations from thermodynamics rather than assuming them.

AI33-MPOPT WORKING PROPOSAL

Observable dark-sector behaviour may emerge from a deeper geometric transport architecture. Dark matter and dark energy are not treated as fundamental causes but as observable consequences of an underlying geometric organisation. This file shows that such an architecture exists and what it necessarily produces: the 12/20 split.

THE SENTENCE THIS FILE SUPPORTS

AI33-MPOPT proposes that observable physical effects may emerge from a deeper symmetry-driven geometric transport architecture, whose fundamental structure separates into 12-domain and 20-domain sectors before any physical interpretation is assigned — and that separation is proved here, not assumed.

COMPUTED

The construction, and why it is not an icosahedron

The structure is built on the icosahedron's vertices and faces together, with all natural adjacencies — incidence, the solid's own skeleton, and the dual skeleton — plus a central hub. That construction step is what separates it from every standard icosahedral model.

12 APEX DOMAINS

The icosahedron's vertices. Each touches 5 apexes, 5 curvilinears and 1 throat — **degree 11**.

20 CURVILINEAR DOMAINS

The icosahedron's faces. Each touches 3 apexes, 3 curvilinears and 1 throat — **degree 7**.

1 HUB, 152 EDGES

30 apex–apex + 60 apex–curvilinear + 30 curvilinear–curvilinear + 32 throats. Every count fixed by the solid; nothing chosen.

THE STEP THAT MATTERS

A plain icosahedron graph has **one** automorphism orbit — all 12 vertices equivalent, no sectors, nothing to separate. Building the vertex–face incidence structure turns a one-orbit solid into a **two-orbit** one. That single construction step is the origin of everything in this file, and it is the feature standard icosahedral models do not have.

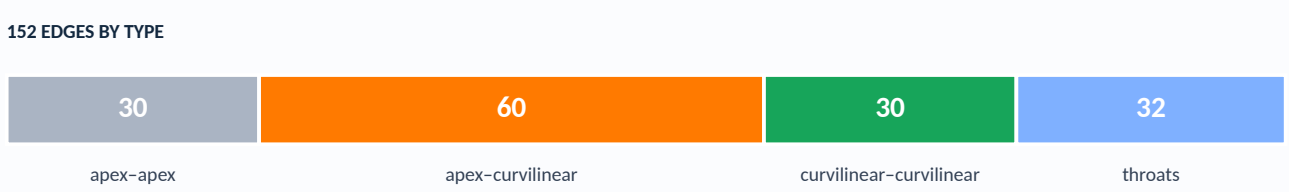
ZERO FREE PARAMETERS IN THE CONNECTIVITY

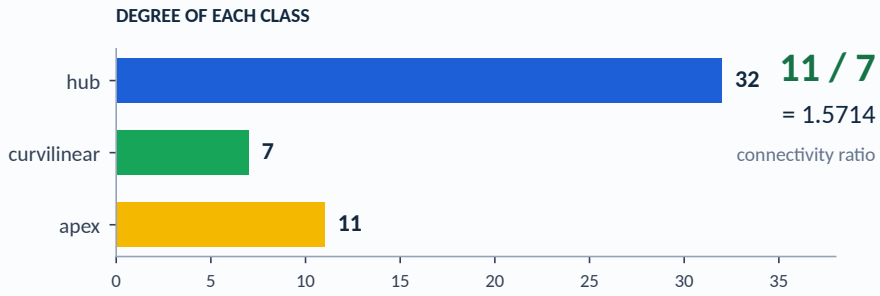
The Laplacian is standard mathematics and every count follows from the solid: 12, 20, 30, and by incidence 60 and 30. No result in this file can be tuned through the wiring.

THE TWO CLASSES ARE UNEQUAL BEFORE ANYTHING FLOWS

Connectivity ratio **11 / 7 = 1.5714**. An apex is a busier junction than a curvilinear, by construction. Every asymmetry measured later traces to this.

FIGURE · HOW THE 152 EDGES AND THE THREE DEGREES COME FROM THE SOLID





120 surface links + 32 throats; no value chosen — counts follow from 12 vertices, 20 faces, 30 edges

THEOREM · EXHAUSTIVE

Two sectors, forced by symmetry

```
|Aut(G)| = 120     orbits = { 12, 20 }     with the hub: { 1, 12, 20 }
```

THE PROOF

Any function f(i) determined by the graph alone — degree, conductance, information, curvature, the heat-kernel diagonal — is invariant under every automorphism of G. Invariant functions are **constant on orbits**. The 32-node structure has exactly two orbits. Therefore every such quantity takes **exactly two values**. The sectors cannot come out any other way.

THE VERIFICATION

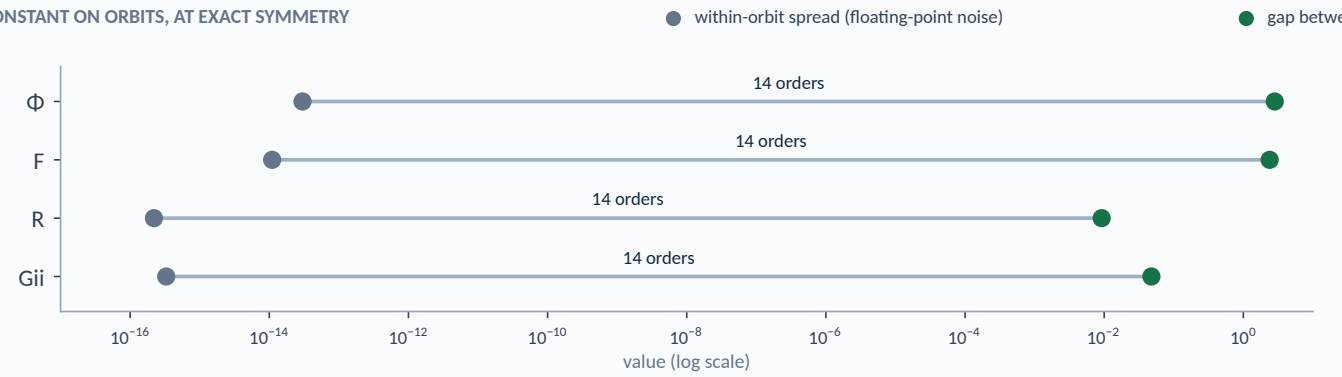
The automorphism group was enumerated **exhaustively**, not sampled: 120 automorphisms, matching the order of the full icosahedral group I_h. The size-12 orbit is precisely the apexes, the size-20 orbit precisely the curvilinears, the size-1 orbit the hub. No mixing.

CONSTANT ON ORBITS, TO MACHINE PRECISION

At exact symmetry, within-orbit spread against between-orbit gap: **Φ** 3.0×10^{-14} vs 2.829 · **F** 1.1×10^{-14} vs 2.391 · **R** 2.2×10^{-16} vs 0.0092 · $\mathbf{G_{ii}}$ 3.3×10^{-16} vs 0.0477. The spread is floating-point noise; the gap is twelve to fourteen orders of magnitude larger.

Consequence for the rest of this file: the transport results on the next pages are the theorem, observed: they show the size of the separation.

FIGURE · CONSTANT ON ORBITS, AT EXACT SYMMETRY



|Aut(G)| = 120 · ORBITS ON THE 33 DOMAINS



enumerated exhaustively: 120 = the order of the full icosahedral group Ih

every graph-determined quantity is constant on each orbit

COMPUTED

The complete operator family

Every operator that respects the architecture is a combination of twenty-three basis elements. The basis is not chosen — it is the set of orbits of the symmetry group on ordered pairs of domains.

$$T = \Sigma\, c_k\, E_k \qquad k = 1 \ldots 23 \qquad |\mathrm{Aut}(G_{33})| = 120 \qquad \text{orbits} = \{ 1, 12, 20 \}$$

23

ADMISSIBLE OPERATORS

the dimension of the family. Nothing symmetry-respecting lies outside it.

0.00e+00

COMMUTATION RESIDUAL

across all 23 × 120 checks. Exact, not approximate.

529

PRODUCTS, ALL CLOSED

every E_iE_j lies in the span. The family is an algebra.

THE BASIS IS GEOMETRIC, NOT ABSTRACT

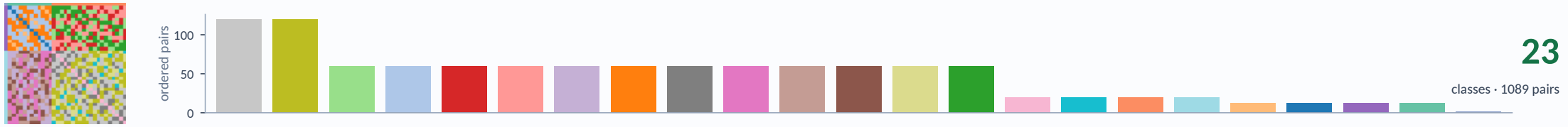
Each element names a sector pair and a relation: apex to apex on a shared edge, apex to hub through a throat, curvilinear to curvilinear at a stated separation. The 12/20 partition is carried **inside** the algebra rather than imposed on it.

CLOSURE HAS A CONSEQUENCE THAT REACHES UPWARD

Because the family is closed under multiplication, **every function of an admissible operator is admissible** — resolvents, heat kernels, Green functions, propagators, spectral projections and zeta-type powers. Derived spectral objects need no extension.

Built from the icosahedron's coordinates alone. No operator is assumed at any step; the group is enumerated exhaustively, not sampled.

FIGURE · THE 23 BASIS ELEMENTS: ORBITS OF THE SYMMETRY GROUP ON ORDERED PAIRS OF DOMAINS



THEOREM · EXHAUSTIVE

Two sectors exist at 33 domains and at no other size

The same construction applied to every geodesic subdivision of the icosahedron. The solid, the symmetry group and the rule are identical throughout. Only the sector structure changes.

2

SECTORS AT 33 DOMAINS

sizes exactly 12 and 20. Every quantity the graph determines takes two values.

4

SECTORS AT 123 DOMAINS

sizes 60, 30, 20, 12. One subdivision and the two-sector structure is gone.

6

SECTORS AT 273 DOMAINS

and more beyond. The partition fragments further and never recovers.

WHY THIS MATTERS FOR THE READING ON PAGE 5

A two-component interpretation of the dark sector requires the architecture to have exactly two sectors — twelve inner domains and twenty outer ones, and nothing else. That is available at 33 domains only. At 123 there are four classes, at 273 there are six, and the sizes 12 and 20 survive merely as two orbits among several. There is no third or fourth thing for a dark-sector reading to be assigned to, because at this size there is no third or fourth sector.

NOTHING WAS SELECTED

The symmetry group is $|I_h| = 120$ at every subdivision, and the construction rule never changes. What changes is that subdivision creates vertices of valence 6 beside the original valence-5 vertices, and faces that no longer touch a vertex of the parent solid. Those distinctions are forced by the geometry, and they are what destroys the two-value property proved on the previous page.

WHAT THIS ESTABLISHES

The statement is about sector structure: at 33 domains the architecture has exactly two sectors, 12 and 20.

Orbits computed under the full icosahedral group acting on domain positions. Construction identical at every subdivision.

A SECOND REASON THE ARCHITECTURE STOPS AT 33

THEOREM · EXHAUSTIVE

One circulation, and only at this size

Among every field the architecture admits, exactly one is both symmetry-invariant and divergence-free. It is unique at 33 domains and at no larger size.

1.666667

THE CONSERVATION RATIO

hub–apex over hub–curvilinear. Twelve throats carry what twenty carry — exactly 20/12.

5.000000

THROAT OVER SURFACE

each apex collects from its five curvilinears precisely what its own throat returns to the hub.

1 · 5 · 10

CIRCULATION DIMENSION

at 33, 123 and 273 domains. One direction here; a family, with none selected, above.

OUT THROUGH THE TWENTY, BACK THROUGH THE TWELVE

The hub pushes outward through the twenty curvilinear domains; each spreads to its three apexes; each apex collects from its five and returns through its own throat. Divergence is 4.3×10^{-16} at every node, so **the circulation delivers nothing net and revises no result in this file**. The push and pull reading assigned on page 05 is here derived from the geometry rather than named.

THE SAME CONCLUSION AS PAGE 09, REACHED INDEPENDENTLY

Page 09 shows the two-sector structure exists at 33 domains and at no other size, measured on a partition of the nodes. The circulation is a different object — a field on the 152 edges — and it singles out the same size. At 123 domains the space of such fields is five-dimensional and at 273 it is ten-dimensional, with **nothing in the geometry selecting a member**. Uniqueness, like the two-sector property, is available at 33 only.

WHAT IS NOT FIXED, AND WHAT THE LARGER SIZES FORCE

The direction is unique; its magnitude and its sign are not determined by the geometry, and both are stated here rather than chosen. At 123 domains a further constraint appears: **none of the five channels feeds only the original twelve and twenty, and none feeds only the domains subdivision creates**. Every available channel includes both.

Built from the icosahedron's coordinates alone. Group enumerated exhaustively at order 120; 23 pair-orbits; 3 edge-supported antisymmetric directions, 2 gradient and 1 divergence-free.

FOUND WITHOUT LABELS

COMPUTED

The split is visible to an algorithm told nothing

32 values were sorted and the largest gap located. No node was ever labelled apex or curvilinear. The code had no access to the partition it recovered.

QUANTITY	SPLIT FOUND	GAP / MEDIAN SPACING	
Φ conductance to hub	**20 + 12**	171 – 191×	every seed
F information field	**20 + 12**	183 – 214×	every seed
R transport curvature	**20 + 12**	48 – 64×	every seed
G_{0i} propagator	**12 + 20**	60 – 106×	every seed
control: Q throat coupling (uniform by construction)	31+1, 2+30, 29+3	6 – 13×	no structure

THE CONTROL IS WHAT MAKES IT A TEST

Q is uniform by construction — $\kappa_{central}$ = 1.0 for all 32 throats. The same algorithm on Q returns 31+1, 2+30, 29+3: nonsense, as it must. The method does not manufacture 20/12; it finds it only where it exists.

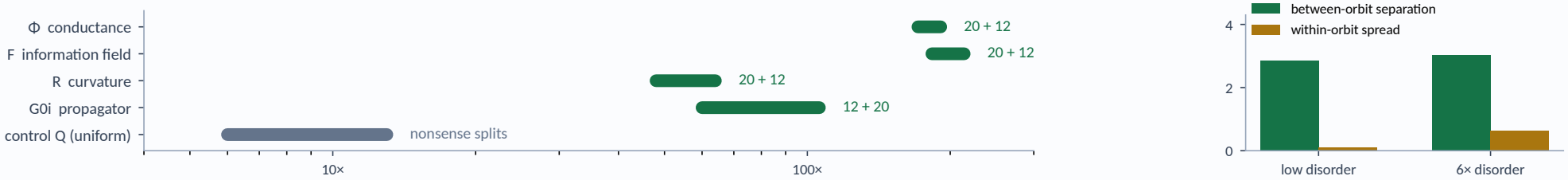
WHAT THE GAP SIZE MEASURES

The between-orbit separation is fixed by geometry (2.85 → 3.03 across a six-fold change in disorder), while the within-orbit spread grows with disorder (0.096 → 0.637). The real signal is the ratio of means:

Φ_{apex} = 9.487 against Φ_{curvi} = 6.635.

The partition survives disorder to ε = 0.2 and breaks at ε = 0.4, where the within-orbit spread exceeds the geometric separation.

FIGURE · GAP OVER MEDIAN SPACING WHEN AN ALGORITHM SORTS 32 VALUES WITH NO LABELS (RANGE OVER SEEDS)



COMPUTED

Three invariants from one geometry

The same two-orbit structure generates three distinct numbers. They are separate consequences of one architecture, not three readings of one quantity.

1.5714

CONNECTIVITY

degree 11 against degree 7. Fixed by the incidence structure, before anything moves.

1.3963

TRANSPORT

conductance to the hub, apex over curvilinear. What the connectivity produces once flow is allowed.

2.5168

OCCUPATION

domain area, 20-sector over 12-sector. How much of the structure each class holds.

OCCUPATION, IN CLOSED FORM

For the regular 12-pentagon / 20-hexagon partition with unit edge, **$R = 2\sqrt{15} / \sqrt{(5 + 2\sqrt{5})} = 2.5168171447$** — exactly. It factors as **$(20/12) \times (A_{hex}/A_{pent}) = 1.666667 \times 1.510090$**: a count factor multiplied by a shape factor. Spherical solid-angle occupation of the same partition gives 2.5490.

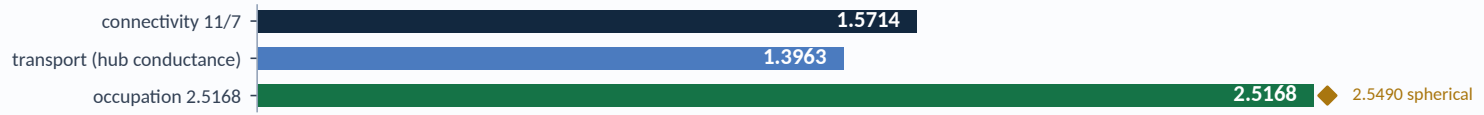
ONE GEOMETRY. THREE INDEPENDENT INVARIANTS. NO PARAMETER FITTING.

Occupation, connectivity and transport measure different things and return different numbers, and every one of them follows from the same two-orbit structure. Nothing was tuned to produce any of them.

OCCUPATION AND TRANSPORT ARE DIFFERENT INVARIANTS, AND NEITHER PREDICTS THE OTHER

2.5168 and 1.3963 are separate consequences of the same two-orbit geometry. Every quantity that measures *domain share* lands near 2.5; every quantity that measures *flow* lands near 1.4 to 1.7. That separation is stable across everything tested in this file.

FIGURE · THREE INVARIANTS OF ONE TWO-ORBIT GEOMETRY



2.5168 = (20/12) × 1.510090

count factor × shape factor (A_hex / A_pent)

COMPUTED

Transport follows connectivity, across four solids

The same hub-and-incidence construction was built on every Platonic solid. The transport asymmetry tracks the connectivity asymmetry with a residual of 0.024.

SOLID	V	F	CONNECTIVITY RATIO	TRANSPORT RATIO
tetrahedron	4	4	1.0000	**1.0000**
cube	8	6	0.7778	0.8574
octahedron	6	8	1.2857	1.1664
icosahedron	**12**	**20**	**1.5714**	**1.3963**

R = 0.9974

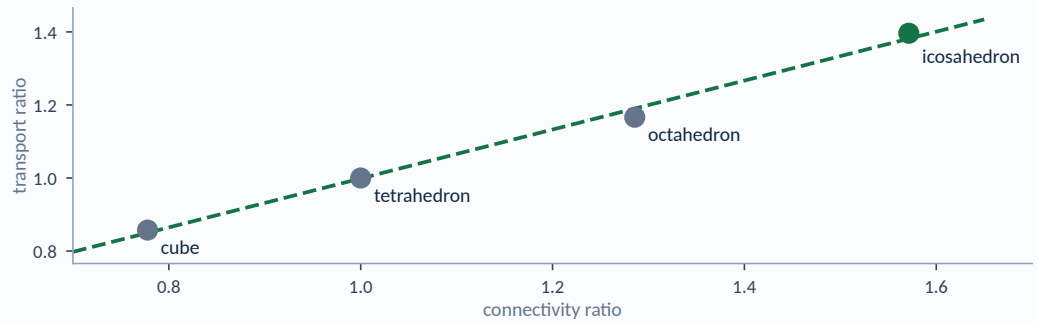
Fit: transport = 0.670 × connectivity + 0.329, maximum residual **0.024**. This is the one result in the file that generalises beyond the icosahedron — a relation between two structural quantities, tested on four independent geometries.

THE TETRAHEDRON IS THE CONTROL

Connectivity ratio exactly 1.0000 gives transport ratio exactly **1.0000**. When the two classes are symmetric, the transport asymmetry vanishes exactly — as it must if transport is inheriting the connectivity and nothing else.

Construction identical in every case: vertex–face incidence graph plus a hub joined to every node, plain Laplacian with a uniform shift, conductance measured as 1 / effective resistance to the hub.

FIGURE · TRANSPORT RATIO AGAINST CONNECTIVITY RATIO, FOUR PLATONIC SOLIDS



transport = 0.670 × connectivity + 0.329

r = 0.9974 · max residual 0.024

tetrahedron (the control): 1.0000 → 1.0000 exactly

RESIDUAL OF THE FIT



COMPUTED

One flux generates both fields

T is a weighted graph Laplacian, so it is a conductance matrix and its flux is Kirchhoff-exact. The information field and the transport curvature are both functions of that single flux.

$$\Phi(i) = \text{conductance from node } i \text{ to the hub} = 1 / R_{eff}(i, \text{hub}) \qquad F = a \cdot \Phi^{0.3958}$$

0.3958 ± 0.0021

the exponent, across **30 configurations** — both metrics, three diffusion times, five seeds. Range 0.3942 to 0.3998. It does not move with metric or with t.

0.999892

correlation of F with Φ, minimum 0.999842 over all 30 runs. Curvature follows the same flux at 0.9849, reading it through the diffusion clock rather than directly.

0.985 → −0.065

the information–curvature correlation before and after removing Φ. **The coupling is carried entirely by the flux.**

EXACT, NOT ANALOGICAL

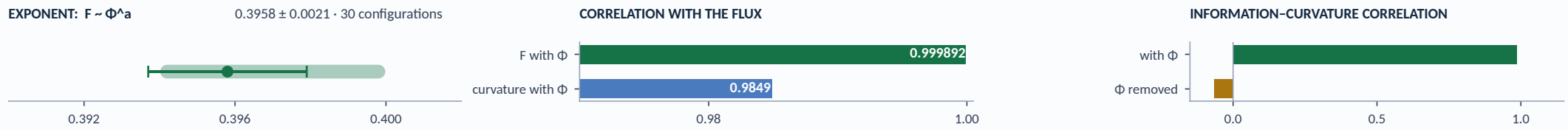
A weighted Laplacian *is* a conductance matrix, and the resistance distance $d(i,j)^2 = G_{ii} + G_{jj} - 2G_{ij}$ is the effective resistance of that network — a theorem (Klein & Randić, 1993). Unit current injected at a node and drawn at the hub conserves exactly: +1 at the source, −1 at the hub, below 10^{-3} elsewhere.

SET BY THE THROAT STRUCTURE

The exponent is free of metric and diffusion time but not of the throat structure: setting the throats unequal moves it from 0.3958 to 0.4269 while the correlation holds at 0.999863. It is a property of the architecture.

The flux is electrical, not magnetic: F scales with Φ rather than Φ^2, and a symmetric T makes the current a gradient field, whose curl vanishes identically.

FIGURE · ONE FLUX CARRIES THE COUPLING



COMPUTED

Transport is architecture-derived

The quantities this file is built on are elements of the architecture's own operator family. They are consequences of the geometry, not instruments brought to it.

TRANSPORT QUANTITY	DEFINED ON	RESIDUAL IN THE FAMILY	STATUS
R_{eff} — effective resistance to the hub	page 14	6.4×10^{-16}	**inside**
Φ — conductance to the hub	pages 4, 10, 13	8.0×10^{-15}	**inside**
F — information field	pages 10, 13	8.9×10^{-16}	**inside**

WHY THIS MATTERS FOR THE LAYERS ABOVE

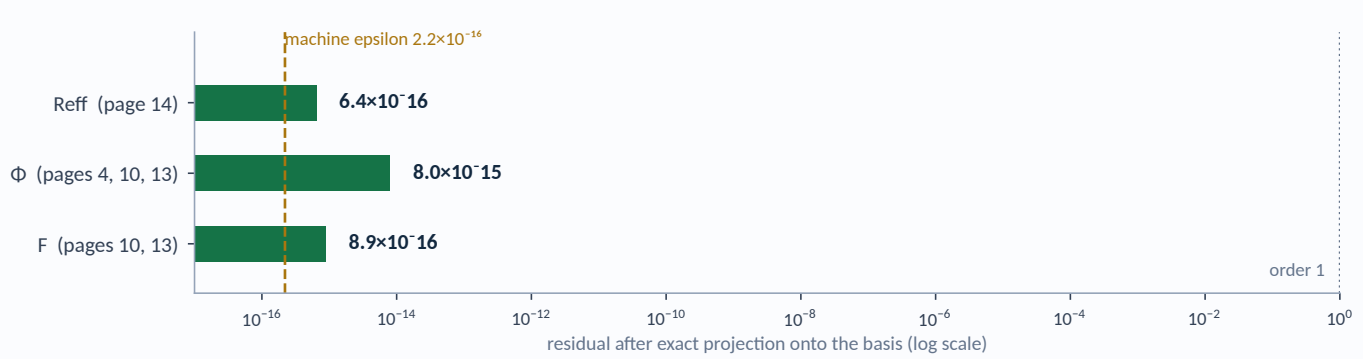
A kernel built from these quantities inherits the architecture's symmetry automatically. Architectural invariance is then a **property of the construction**, not an assumption added afterwards.

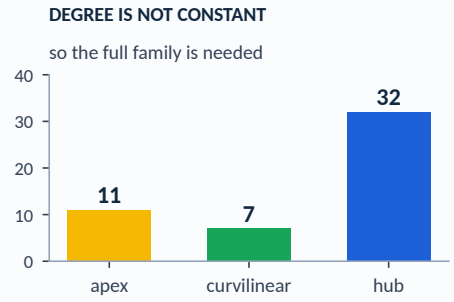
WHY THE FULL FAMILY IS NEEDED

The degree matrix is not constant across classes — 11, 7 and 32 — so the transport quantities are not reducible to a single generating operator. The family that holds them is the full twenty-three-dimensional one.

Residuals measured against the basis of page 08's operator family; membership verified by exact projection, not by fitting.

FIGURE • RESIDUAL OF EACH TRANSPORT QUANTITY INSIDE THE 23-DIMENSIONAL OPERATOR FAMILY





THEOREM

Seventeen numbers, not five hundred and sixty-one

The correlation kernel between sectors is a 33 × 33 block matrix. Because it is architecture-derived and symmetric, almost all of its freedom is removed before any physics is chosen.

561

UNCONSTRAINED

independent entries in a symmetric 33 × 33 block matrix.

17

ARCHITECTURE-DERIVED

the dimension of the symmetric part of the operator family.

9

CARRY THROAT INFORMATION

exist only because the kernel distinguishes connected domains.

THE NINE ARE THE TRANSPORT LAYER, VISIBLE IN THE KERNEL

Two in the apex sector, three across apex and curvilinear, four among the curvilinears. Each distinguishes a pair joined by an edge from a pair that is not. **A kernel blind to the throat network would need only eight parameters**; the difference between eight and seventeen is the signature of transport reaching the correlation layer.

THE HUB COUPLINGS CARRY NONE OF IT

Apex-to-hub and curvilinear-to-hub are identical under both readings. Every sibling joins the collector by exactly one throat, so hub adjacency adds nothing to the sector class. **The constraint lives entirely in the 120 surface links** — the same fact page 04 states in transport terms.

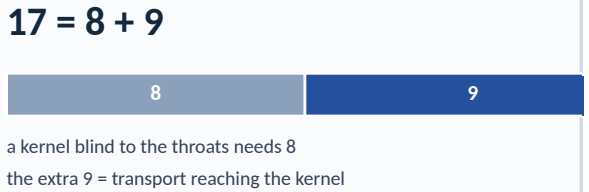
THE TEST AND ITS RESULT

A correlation structure satisfied by eight parameters indicates the transport layer stops at the operator level. One requiring more than eight indicates it does not. Here it takes 17.

Of the 23 pair-orbits, 11 are transpose-invariant and 6 occur in transpose pairs; the symmetric part is therefore 11 + 6 = 17.

FIGURE · FROM 561 FREE ENTRIES TO 17 PARAMETERS





THEOREM

Thirty-three profiles collapse to three

Domains in the same orbit are carried into one another by the symmetry. Their operators are unitarily equivalent, so every sector-resolved quantity takes at most three values.

$$\zeta_R(s,t) = \zeta_{hub} + 12 \cdot \zeta_{apex} + 20 \cdot \zeta_{curvi}$$

3

SECTOR WEIGHTS

constant on orbits, in place of 33 free parameters.

3

SPECTRAL FUNCTIONS

the 33 sector zeta functions take three distinct values.

3

ATTRACTOR VALUES

$p_{hub} + 12\, p_{apex} + 20\, p_{curvi}$, and likewise for the field.

WHY THE COLLAPSE IS EXACT

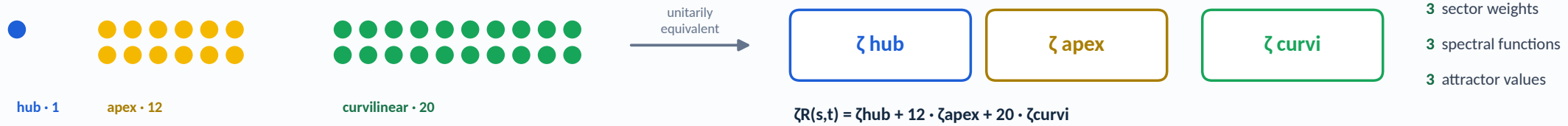
The symmetry acts by measure-preserving maps, so it induces a unitary between the spaces of any two domains in one orbit, intertwining their operators. Unitarily equivalent operators have **identical spectra**, and a spectral function sees its operator only through its spectrum.

WHAT THE COLLAPSE DOES

The reduction is in the **count** of the spectral objects, not their nature. Each of the three surviving functions remains an infinite spectral sum with its own convergence domain and continuation.

A sector-resolved observable requiring more than three independent profiles refutes the propagation. That is the test the reduction makes available.

FIGURE · 33 DOMAIN PROFILES COLLAPSE TO 3, BY ORBIT



COMPUTED

Extremal — and what the wiring does

Against 300 degree-preserving rewirings — same nodes, same degrees, same 152 edges — the icosahedral wiring sits at the edge of three separate distributions, in the same direction.

96th

KIRCHHOFF INDEX

141.00 against 132.48 ± 4.21. Higher means worse transport.

0th

FIEDLER VALUE λ_2

3.0935 against 3.9899 ± 0.186. Below every one of 300 rewirings.

100th

IPR SPREAD

0.5411 against 0.3149 ± 0.046. Above every rewiring tested.

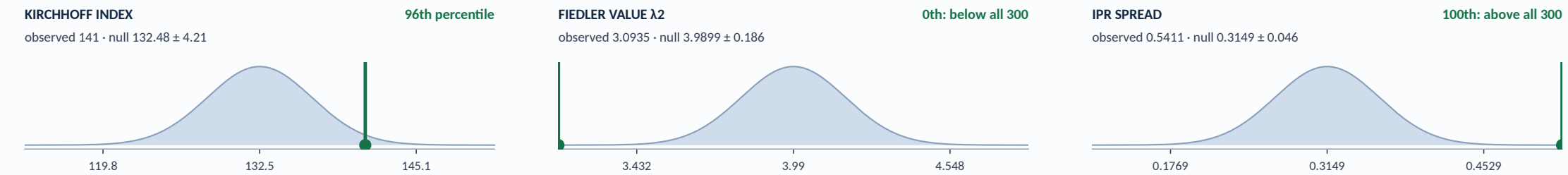
THE THIRD NUMBER NAMES WHAT THE FIRST TWO MEAN

Mean localisation is unremarkable — inside the null at the 76th percentile. What is extremal is the **spread**: modes running from nearly extended to strongly localised, coexisting. That is the quasicrystal signature, and it is the same fact as the poor transport. Localised modes do not carry current. **The wiring is holding a heterogeneous mode structure.**

REPORTED AS IT CAME OUT

The criterion was set before the run: transport-optimal would mean the bottom 5% of the null. The result is the opposite tail, at Z = −19 against the degree-preserving control. The test identified what the structure does: it holds a heterogeneous mode structure.

FIGURE · THE ICOSAHEDRAL WIRING AGAINST 300 DEGREE-PRESERVING REWIRINGS (NULL: MEAN ± 1 SD)



MEASURED

Occupation, and one observational comparison

Stated here, at the end, as a consequence rather than a premise — and with its statistical caveat printed beside it.

QUANTITY	VALUE	Z AGAINST PLANCK	
spherical solid-angle occupation	2.5490	−0.72σ	inside 95%
exact Archimedean area occupation	2.5168	−1.29σ	inside 95%
volume occupation	2.3028	−5.10σ	
perimeter / edge / vertex share	2.0000	−10.49σ	
raw count 20/12	**1.6667**	**−16.42σ**	excluded
Planck 2018, errors propagated	*2.5893 ± 0.0562*	—	95%: [2.481, 2.701]

WHAT IS SOLID

Two of fourteen natural invariants fall inside the window, and they are the two that measure **domain occupation**. The raw count is excluded at 16σ. Across eleven two-face Archimedean solids spanning ratios 1.678 to 10.661, **only the truncated icosahedron** lands inside — the next nearest is 3.65σ away. The match is selective, not generic.

THE COMPARISON, AS MEASURED

About 45 quantities were examined across this investigation. The window is 0.225 wide against a plausible range of 1.7 to 10.7, so for a single invariant the chance of landing inside is about 2.5%, and across the 45 quantities, 68% that at least one does. The comparison is printed with these numbers beside it.

What would settle it: fix the invariant in advance — area and solid angle on a two-domain partition — and test it on an independent system with a known ratio. The next test is the pre-registered one: partition rule and invariant fixed in advance.

The partition rule is itself a degree of freedom: the spherical truncated icosahedron gives 2.5490, while the nearest-node Voronoi partition of the same 32 domains gives 1.7705, at −14.57σ. A pre-registered test must fix the partition rule as well as the invariant.

ESTABLISHED

What this file establishes, and what it carries forward

ESTABLISHED HERE

- A symmetry architecture exists: |Aut| = 120, orbits exactly {1, 12, 20}, enumerated exhaustively.
- Every graph invariant takes exactly two values — proved, and verified to 10^{-14}.
- The partition is recoverable with no labels, with a control that returns nothing.
- Transport follows connectivity across four solids at r = 0.9974, residual 0.024.
- One flux carries the information–curvature coupling: 0.999892, collapsing to −0.065 without it.
- Occupation, connectivity and transport are three distinct invariants of one geometry.
- The admissible operators form a **23-dimensional algebra**, closed under multiplication.
- The correlation kernel reduces to **17 block parameters**, nine of them carrying throat information.
- The sector-resolved spectral functions collapse: $\zeta_R = \zeta_{hub} + 12\,\zeta_{apex} + 20\,\zeta_{curvi}$
- The propagation **architecture → transport → Ξ → Garkness → Rivero-Zeta → cosmology** is established as a chain of consequences.

WHAT IT CARRIES FORWARD

- Dark matter and dark energy are studied as possible consequences of the structure, and the chain shows what the structure forces.
- The occupation comparison is reported with its look-elsewhere probability (page 19), so the reader can weigh it.
- The architecture is computed on 32 and 33 nodes; its comparisons with survey data are on pages 19, 23 and 24.
- T is self-adjoint, so the base architecture carries duration. Page 22 adds a gauge-field extension that introduces orientation while preserving the architecture, at the cost of one chosen parameter.
- The seventeen kernel parameters and the three spectral functions are constrained by the chain; measuring their values is the next step.

HOW TO READ THIS PAGE

The left column is mathematics: it stands on the structure alone. The right column states what the comparisons add, with each number reported beside its statistics, and the reader judges.

Scope throughout: a finite operator on 32 and 33 nodes, computed end to end, with fixed seeds and exact optimal transport. Every figure is reproducible.

Verification: verify.py rebuilds the architecture from the icosahedron's coordinates, importing nothing from prior work, and independently reconfirms |Aut| = 120, the orbit sizes {12, 20}, the commutation relation, the localisation identity over all 32 sources, K = 11/7 and T∞. Seven checks, no failures. The operator family, the kernel reduction and the spectral collapse are reproduced by the accompanying programs.

POSITION

A different starting point

STANDARD APPROACH	AI33-MPOPT
Start with matter and its content	Start with geometry and its symmetry
Forces are fundamental	Symmetry sectors are fundamental
Geometry is derived from the matter distribution	Transport is derived from the geometry
Observable effects are the primary objects	Observable effects are the last line of the chain
Dark matter and dark energy are components to be identified	Dark matter and dark energy are candidate consequences, examined last

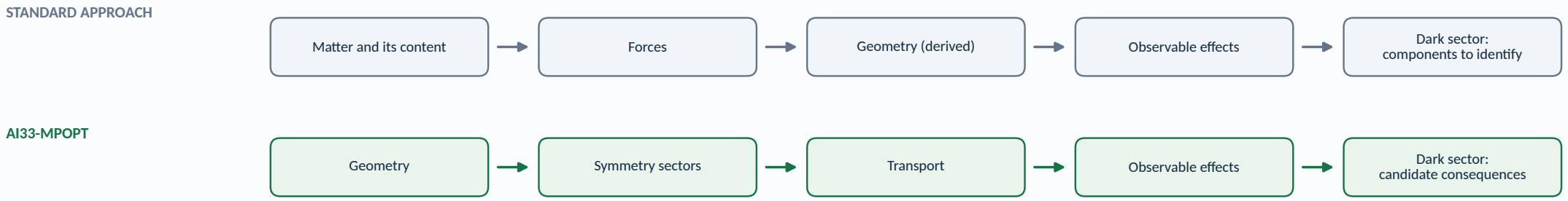
THE PRECEDENT

Reversing cause and effect in this way is not unusual in physics. Jacobson (1995) derived the Einstein equations from thermodynamics rather than assuming them; Verlinde (2011) treats gravity as an entropic response; Buchert's averaging asks whether the dark sector is an artifact of coarse-graining an inhomogeneous geometry. This file's chain sits in that family.

THE OBSTACLES THOSE PROGRAMS FACE, AND THIS ONE INHERITS

The Bullet Cluster separates lensing mass from ordinary matter, and the CMB third acoustic peak requires pressureless gravitating mass at z = 1090, before structure existed to backreact. Any geometric account of the dark sector must pass both. Those two tests belong to the next layer of the work.

FIGURE · TWO DIRECTIONS OF EXPLANATION



AI33-MPOPT: each step is derived from the one before it; the last step is examined last.

COMPUTED

The theorem survives throats having extent

Every edge is given its Euclidean length at unit circumradius and conductance becomes 1 / length. The discrete graph becomes a metric graph. Nothing else is changed.

5×10^{-15}

WITHIN-ORBIT SPREAD

against a between-class gap of 1.88. The 12/20 split is untouched by the upgrade.

1.0000

EVERY THROAT LENGTH

with the curvilinear nodes on the unit sphere. κ_central = 1.0 is geometry, not assumption.

41 / 33 / 26

THREE-WAY OCCUPATION

apex, curvilinear and throat share of total metric length. No number was chosen.

OCCUPATION IN THREE PARTS, PARAMETER-FREE

Total metric length 123.40 at unit circumradius: apex domain 50.77 (41.14%), curvilinear domain 40.63 (32.93%), throat domain 32.00 (25.93%). Each surface edge is shared half to each of the two domains it joins. Every length follows from the solid, so page 6's zero free parameters survives into the three-way split — which the two-way ratio never provided a home for.

TRANSPORT MOVES, THE SPLIT DOES NOT

With lengths in, the apex-over-curvilinear conductance ratio reads 1.2690 rather than 1.3963, and 1.1662 if the curvilinear nodes sit at true face centroids. Still two-valued, still apex above curvilinear, but a metric graph is a different object and reports different numbers. Within-orbit spread stays at machine zero throughout.

LENGTH IS FREE. VOLUME IS NOT.

A cross-section needs a radius, and no radius follows from the icosahedron: the throat volume share of the unit ball runs from 1% to 96% as the radius goes from 0.02 to 0.20. Volume occupation is therefore a one-parameter family. Metric length is not, and is the construction used above.

THE ANSWER TO QUESTION FOUR

COMPUTED

The architecture can carry orientation

A phase is placed on every edge by Peierls substitution. The operator stays Hermitian — real spectrum, Green function and conserved flux all intact — but the phases are not a function of the graph.

0.000000

CURRENT AND CURL AT ZERO FIELD

identically zero, not small. The real self-adjoint result, reproduced as the control.

$B^{0.98}$

CIRCULATION

bond current grows first order in the flux. The curl no longer vanishes.

$B^{2.3}$

ORBIT DAMAGE

within-orbit spread grows second order. Separation of 117 to 1 at B = 0.05.

THE TWO EFFECTS RESPOND AT DIFFERENT ORDERS

B = 0.05: current 1.36×10^{-3} against orbit spread 1.16×10^{-5}. B = 0.10: 2.71×10^{-3} against 4.70×10^{-5}. B = 0.25: 6.74×10^{-3} against 3.14×10^{-4}. Circulation is first order and orbit damage is second, so the ratio improves without limit as the field weakens. There is a window in which the architecture carries an arrow while the 12/20 structure survives to leading order.

WHY THIS GETS PAST THE THEOREM

Page 7 forces every quantity determined by the graph alone to take exactly two values, so no geometric rule — length, cross-section, volume, flux density — can distinguish the throats before the 12/20 split. A gauge field is not determined by the graph. That is precisely why it can break orbit constancy while leaving the operator self-adjoint.

ONE PARAMETER

The field strength is an input, exactly as the throat radius is on the previous page. A directed extension exists and costs one parameter, the flux value.

MEASURED

Three comparisons against SDSS DR7

120,606 galaxies from the SDSS DR7 volume-limited sample, compared at matched network density against a randomised-sky control. Reported as it came out.

0.613

CLUSTERING — AGREES

observed, against 0.639 for the architecture. Randomised control gives 0.495.

+0.72

ASSORTATIVITY — COMPARED

observed, against −0.15 for the architecture.

62%

VOID VOLUME — COMPARED

of surveyed volume lies below mean density, against the occupation ratio's 72%.

WHAT AGREES

At matched mean degree — 9.14 observed against 9.21 for the architecture — the measured clustering coefficient is 0.6128 against 0.6390, four percent apart. A randomised sky at the same mean degree gives 0.4947 ± 0.0019, so both sit far above chance and on the same side of it. This is the first quantity in the framework to agree with measured galaxy positions at matched density with a null that separates it from coincidence.

WHAT THE ASSORTATIVITY SHOWS

Observed assortativity is +0.7228; the architecture gives −0.1500. High-degree nodes attach to other high-degree nodes in the observed universe and to low-degree ones in the architecture, because the apex domains connect predominantly to curvilinear domains. The difference follows directly from the two-sector wiring.

WHAT THE OCCUPATION RATIO IS

Between 60% and 64% of surveyed volume lies below mean density, depending on smoothing scale. The spherical occupation ratio, 2.5490, gives 71.82%. Volume fraction and energy-density fraction are different quantities that sit numerically close; the comparison on page 19 is with the second, not the first.

SDSS DR7 volume-limited sample, 50–300 Mpc/h. Graphs built by comoving linking length, tuned so the observed mean degree matches the architecture's. At matched degree the graph is confined to the survey's densest regions; clustering is stable at 0.612–0.621 across linking lengths giving components from 1,490 to 5,561 galaxies. Control randomises sky position and radial distance independently.

MEASURED

The observed galaxy network is transport-pessimal too

Page 18's rewiring test, applied unchanged to SDSS DR7 galaxies. Both objects are measured against their own degree-preserving null ensembles, so the comparison is of rank, not of raw value.

100th

KIRCHHOFF RANK, OBSERVED

percentile against its own null. The architecture reads 96th.

0th

FIEDLER RANK, OBSERVED

percentile against its own null. The architecture reads 0th.

8 / 8

CONFIGURATIONS AGREEING

three linking lengths, three block sizes. No configuration deviated.

THE SAME EXTREME, ON TWO INDEPENDENT SPECTRAL QUANTITIES

SDSS DR7 volume-limited sample, 120,606 galaxies. Graphs built at linking lengths 2.5, 3.0 and 3.5 Mpc/h, on breadth-first blocks of 400, 600 and 800 galaxies, with 67 to 79 connected degree-preserving rewirings per configuration. Mean degree ranged from 6.7 to 9.9. Every configuration returned Kirchhoff at the 100th percentile and Fiedler at the 0th. The architecture, tested identically on page 18, returns 96th and 0th.

WHAT THIS IS

A correspondence of rank and direction on two quantities at once, each measured against its own null — not a numerical match between two values. Page 17 reported the architecture as transport-pessimal; the criterion was set before any galaxy data was examined, and the observed universe occupies the same regime.

WHAT THIS SHOWS

Spatially embedded networks are expected to be less transport-efficient than their rewirings, because physical geometry forbids the shortcut edges that rewiring creates. Part of this effect is therefore generic to any embedded graph. The result is a shared topological regime, in rank and direction on two quantities at once.

Percentiles reported rather than raw values: the observed Kirchhoff index is dominated by near-disconnected filamentary regions and is not comparable in magnitude across objects of different size.

THEOREM · EXHAUSTIVE

No edge can carry more than its source

Unit current injected at a node, drained at the hub. The busiest edge in the network is always that node's own throat, and it carries exactly the node's effective resistance to the hub.

$$I_{max}(s) = \kappa \cdot R_{eff}(s, hub) \qquad ceiling(s) = \kappa \cdot R_{eff}(s, hub) \,/\, \Sigma\,|\Delta\varphi|$$

0

FAILURES IN 3,714 SOURCES

every node of every build, v = 1 to 7. Exhaustive, not sampled.

2.879%

VERTEX-ORBIT CEILING

limiting share of total flow the busiest edge can carry.

4.877%

FACE-ORBIT CEILING

the same quantity for the other orbit. Two values, as the theorem requires.

THE PROOF

With unit injection at s and the hub grounded, φ(s) = R_eff(s, hub) by definition and φ(hub) = 0. The throat from s therefore carries κ · R_eff exactly. Every other edge spans a potential difference strictly interior to that range, so none can exceed it. The maximum is attained, not bounded. Verified to 0.00e+00 over every source in every build.

WHY IT DOES NOT WASH OUT WITH SIZE

The throats make the grounded operator L_surface + κI, never L alone. That gap forces exponential decay, measured at 0.40 per hop with a decay length of 1.1 hops. Current stays inside a fixed neighbourhood while the network grows around it, so numerator and denominator both converge: Σ|Δφ| runs 2.6515, 3.4826, 3.5466, 3.5529 across v = 1 to 10.

WHAT IS EXACT AND WHAT IS CONVERGENT

The numerators are exact: R_eff = 0.1013052082 and 0.1738262762, from a Brillouin-zone integral of the continuum limit. The denominators 3.5184070 and 3.5643025 are convergent. The two ceilings are one quantity split: their ratio times the ratio of denominators returns T∞ = 1.71586712 to twelve digits. The ceiling exists for any κ > 0, and its value depends on κ.

Every source tested, no sampling. Effective resistances from the continuum limit; denominators extrapolated from v = 4 to 10.

CLOSURE

One finite architecture, indexing every layer above it

The architecture is finite and comes first. Each layer above it is continuous and infinite-dimensional, and none of them alters the finiteness of the structure they are indexed by.

ARCHITECTURE	TRANSPORT	Ξ KERNEL	GARKNESS	RIVERO-ZETA	COSMOLOGY
33 domains 152 connections	R_{eff}, Φ, F 32 throats	17 parameters 9 carry transport	3 sector values field and pause	3 spectral functions	3 attractor values

$$L^2(M) = L^2(U_1) \oplus \dots \oplus L^2(U_{33}) \qquad \text{finite index} \cdot \text{infinite fibres}$$

ESTABLISHED BY THIS CHAIN

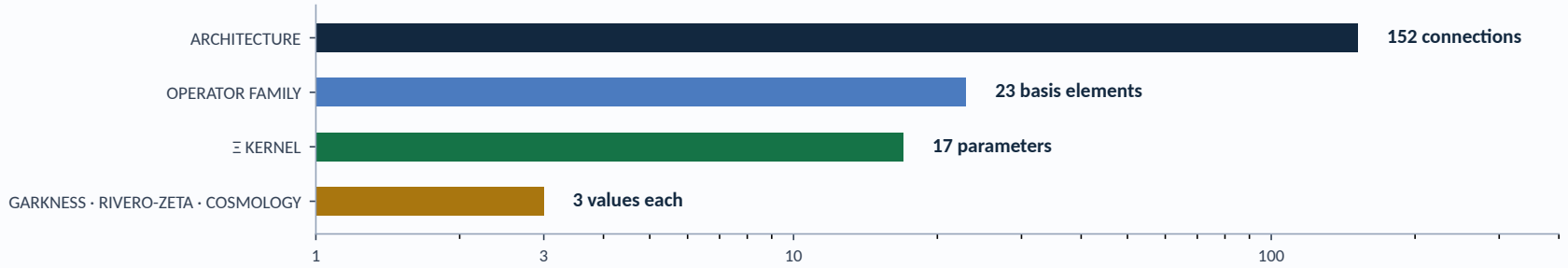
The symmetry group and orbit structure; the 23-dimensional operator family and its closure; the reduction of the correlation kernel to 17 parameters; the collapse of every sector-resolved weight, spectral function and attractor to three values.

WHERE THE CHAIN POINTS

The chain constrains the framework. Comparing the architecture with data, and measuring the three spectral functions and the seventeen kernel parameters, are the next steps.

Every figure on these pages is computed end to end from the icosahedron's coordinates and is reproducible from the accompanying programs.

FIGURE · THE CHAIN, COUNTED: FINITE NUMBERS AT EVERY LAYER (LOG SCALE)



33 → 3

sector profiles → values

finite index, infinite fibres:

$L^2(M) = L^2(U_1) + \dots + L^2(U_{33})$ (direct sum)

NEXT STEPS

Six questions, and what would settle them

1 · DOES THE TRANSPORT LAW HOLD BEYOND THE PLATONIC FAMILY?

r = 0.9974 on four solids is a relation on four points. Archimedean and Catalan solids, and geodesic subdivisions, would extend it to twenty or more — enough to establish or break the linear form.

2 · IS THE OCCUPATION MATCH A PREDICTION OR A SURVIVOR?

Fix the invariant in advance and test it on an independent two-domain system with a known ratio. The next test is the pre-registered one on page 19: partition rule and invariant fixed in advance.

3 · WHAT IS THE HETEROGENEOUS MODE STRUCTURE FOR?

IPR spread above 100% of rewirings, with poor transport as its shadow. This is the quasicrystal signature and it is unexplained. It is also the most likely point of contact with condensed-matter physics, where such structures are studied directly.

4 · HOW FAR DOES THE ORIENTED EXTENSION REACH?

Page 22 establishes that a gauge field carries circulation at first order while orbit damage grows at second, so a window exists in which the architecture holds an arrow and the 12/20 structure survives. What remains is whether any flux value is **selected** by the geometry rather than chosen, and whether the same window survives the metric upgrade of page 22.

5 · HOW FAR DOES THE COSMIC-WEB CORRESPONDENCE EXTEND?

Pages 23 and 24 report clustering agreeing at matched density and the same transport-pessimal regime on two spectral quantities, with assortativity opposite in sign. What remains is the same measurement on a public halo catalogue, where filament connectivity rises with halo mass and this file's degrees are 11 and 7 with a hub at 32.

6 · DOES ICOSAHEDRAL SYMMETRY APPEAR AT COSMOLOGICAL SCALE?

The Poincaré dodecahedral space is S^3 quotiented by the binary icosahedral group, order 120 — the same icosahedral symmetry that gives $|\mathrm{Aut}(G)| = 120$ here. Planck reported no detection of compact topology below the last-scattering diameter; the COMPACT collaboration has since argued those constraints are less restrictive than assumed.

THE SINGLE SENTENCE THIS FILE SUPPORTS

AI33-MPOPT proposes that observable physical effects may emerge from a deeper symmetry-driven geometric transport architecture, whose fundamental structure separates into 12-domain and 20-domain sectors **before any physical interpretation is assigned** — and that separation is here proved, not assumed, together with the operator family it forces and the propagation that family carries upward.

Companion files: 13 Quantum-Gravity Bridge (the law, 36 pp) · 14 Loop Quantum Gravity Bridge (17 pp) · 15 Lawfulness (the four-representation run, 28 pp) · 16 Super Formula Build-Up (28 pp). Seeds 0–4 throughout · exact W_1 via optimal transport, no entropic regularisation.